Interest Rates and Money Markets

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Preview

- What is Money?
- Control of the Supply of Money
- The Willingness to Hold Monetary Assets
- A Model of Real Monetary Assets and Interest Rates

What Is Money?

- Money is an asset that is widely used as a means of payment.
  - Different groups of assets may be classified as money.
    - Money can be defined narrowly or broadly.
    - Currency in circulation, checking deposits, and debit card accounts form a narrow definition of money.
    - Deposits of currency are excluded from this narrow definition, although they may act a substitute for money in a broader definition.

What Is Money? (cont.)

- Money is a *liquid* asset: it can be easily used to pay for goods and services or to repay debt without substantial transaction costs.
  - But monetary or liquid assets earn *little or no interest*.
- Illiquid assets require substantial transaction costs in terms of time, effort or fees to convert them to funds for payment.
  - But they generally earn a higher interest rate or rate of return than monetary assets.

What Is Money? (cont.)

- Let's group assets into monetary assets (or liquid assets) and non-monetary assets (or illiquid assets).
- The demarcation between the two is arbitrary,
  - but currency in circulation, checking deposits, debit card accounts, savings deposits, time deposits are generally more liquid than bonds, loans, deposits of currency in the foreign exchange markets, stocks, real estate, and other assets.

Money Supply

- The central bank substantially controls the quantity of money that circulates in an economy, the **money supply**.
  - In the US, the central banking system is the Federal Reserve System.
    - The Federal Reserve System directly regulates the amount of currency in circulation.
    - It indirectly influences the amount of checking deposits, debit card accounts, and other monetary assets.

Money Demand

- **Money demand** represents the amount of monetary assets that people are willing to hold (instead of illiquid assets).
  - What influences willingness to hold monetary assets?
  - We consider individual demand of money and aggregate demand of money

What Influences Demand of Money for Individuals and Institutions?

1. **Interest rates/expected rates of return** on monetary assets relative to the expected rates of returns on non-monetary assets.
2. **Risk**: the risk of holding monetary assets principally comes from unexpected inflation, which reduces the purchasing power of money.
   - but many other assets have this risk too, so this risk is not very important in defining the demand of monetary assets versus non-monetary assets
3. **Liquidity**: A need for greater liquidity occurs when the price of transactions increases or the quantity of goods bought in transactions increases.

What Influences Aggregate Demand of Money?

1. **Interest rates/expected rates of return**: monetary assets pay little or no interest, so the interest rate on non-monetary assets like bonds, loans, and deposits is the opportunity cost of holding monetary assets.
   - A higher interest rate means a higher opportunity cost of holding monetary assets → lower demand of money.
2. **Prices**: the prices of goods and services bought in transactions will influence the willingness to hold money to conduct those transactions.
   - A higher level of average prices means a greater need for liquidity to buy the same amount of goods and services → higher demand of money.

What Influences Aggregate Demand of Money? (cont.)

3. **Income**: greater income implies more goods and services can be bought, so that more money is needed to conduct transactions.
   - A higher real national income (GNP) means more goods and services are being produced and bought in transactions, increasing the need for liquidity $\rightarrow$ higher demand of money.

A Model of Aggregate Money Demand

The aggregate demand of money can be expressed as:

$$M^d = P \times L(R, Y)$$

where:

P is the price level

Y is real national income

R is a measure of interest rates on non-monetary assets

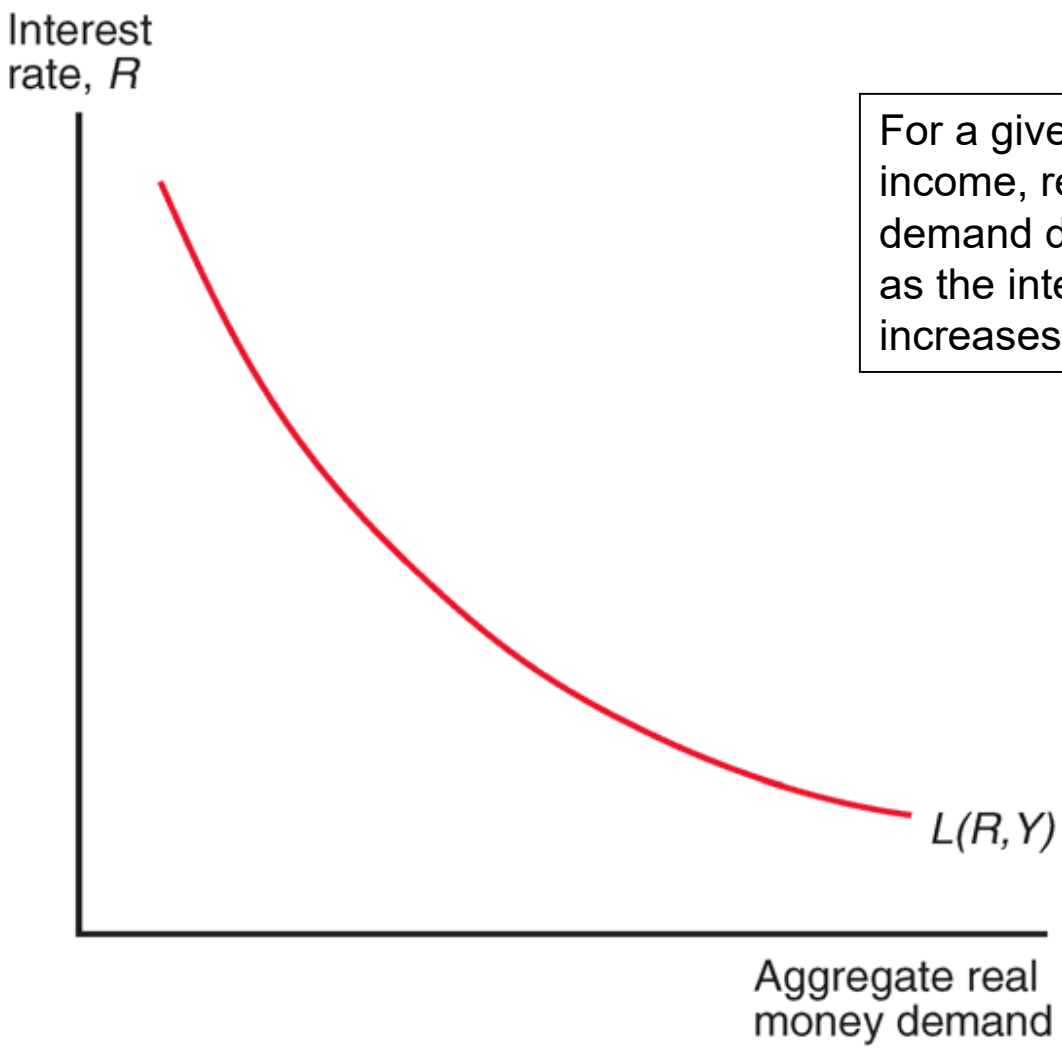
$L(R, Y)$ is the aggregate demand of real monetary assets

Alternatively:

$$M^d/P = L(R, Y)$$

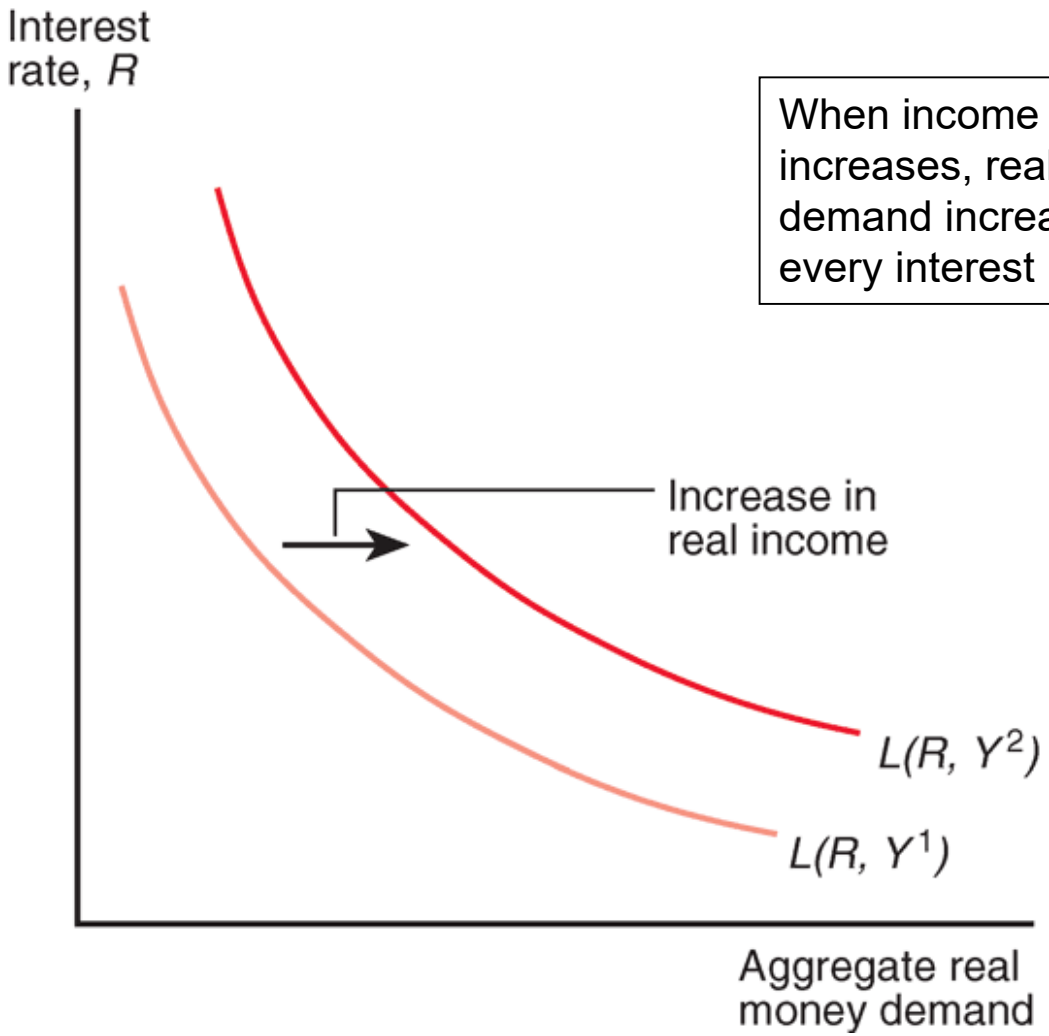
Aggregate demand of real monetary assets is a function of national income and interest rates.

Fig. 1: Aggregate Real Money Demand and the Interest Rate



For a given level of income, real money demand decreases as the interest rate increases.

Fig. 2: Effect on the Aggregate Real Money Demand Schedule of a Rise in Real Income



When income increases, real money demand increases at every interest rate.

A Model of the Money Market

- The money market is where monetary or liquid assets, which are loosely called “money,” are lent and borrowed.
  - Monetary assets in the money market generally have low interest rates compared to interest rates on bonds, loans, and deposits of currency in the foreign exchange markets.
  - Domestic interest rates directly affect rates of return on domestic currency deposits in the foreign exchange markets.

A Model of the Money Market

- When no shortages (excess demand) or surpluses (excess supply) of monetary assets exist, the model achieves an equilibrium:

$$M^s = M^d$$

- Alternatively, when the quantity of real monetary assets supplied matches the quantity of real monetary assets demanded, the model achieves an equilibrium:

$$M^s/P = L(R, Y)$$

A Model of the Money Market (cont.)

- When there is an excess supply of monetary assets, there is an excess demand for interest bearing assets like bonds, loans, and deposits.
  - People with an excess supply of monetary assets are willing to offer or accept interest-bearing assets (by giving up their money) at lower interest rates.
  - Others are more willing to hold additional monetary assets as interest rates (the opportunity cost of holding monetary assets) falls.

A Model of the Money Market (cont.)

- When there is an excess demand of monetary assets, there is an excess supply of interest bearing assets like bonds, loans, and deposits.
  - People who desire monetary assets but do not have access to them are willing to sell non-monetary assets in return for the monetary assets that they desire.
  - Those with monetary assets are more willing to give them up in return for interest-bearing assets as interest rates (the opportunity cost of holding money) rises.

Fig 3: Determination of the Equilibrium Interest Rate

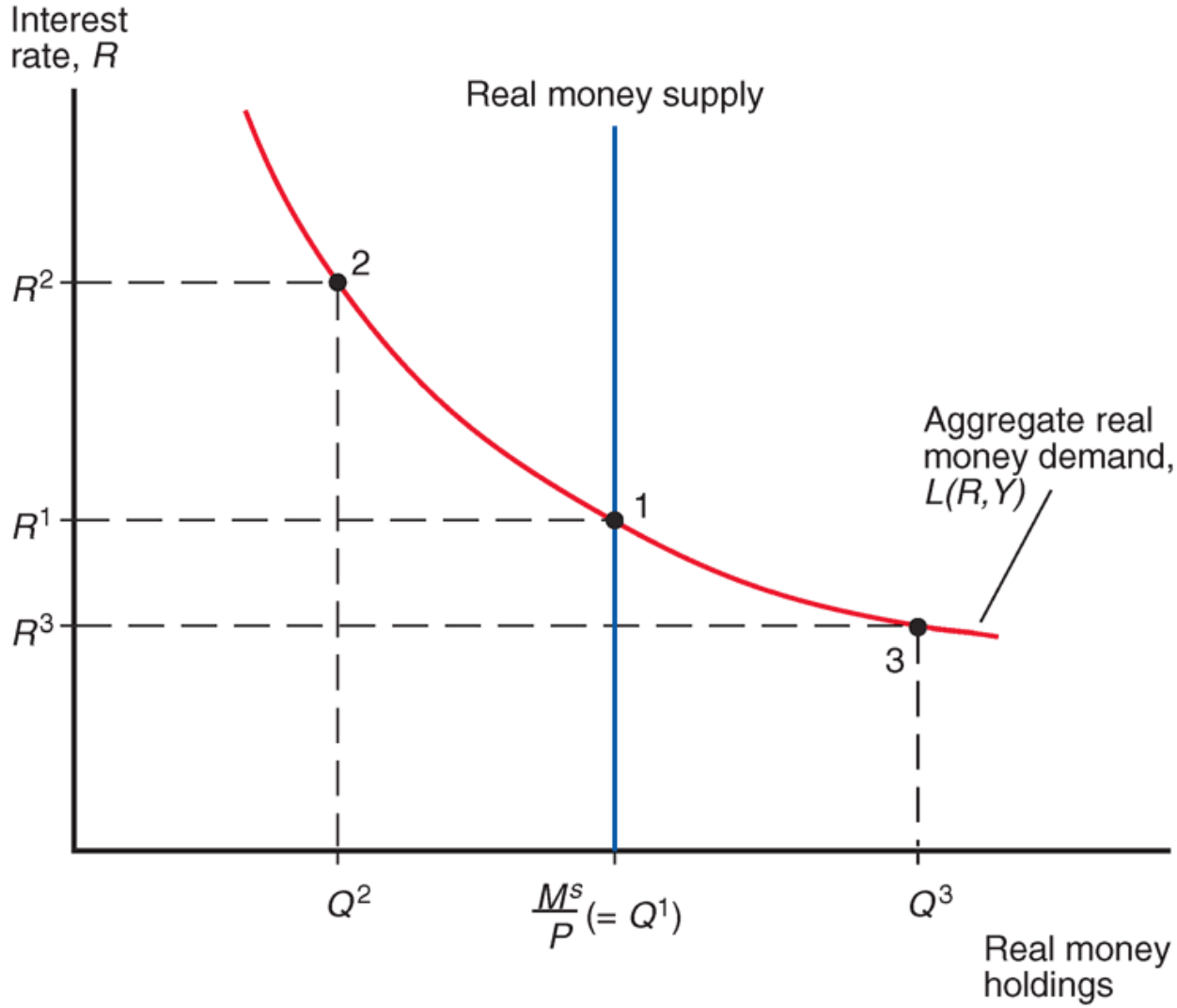
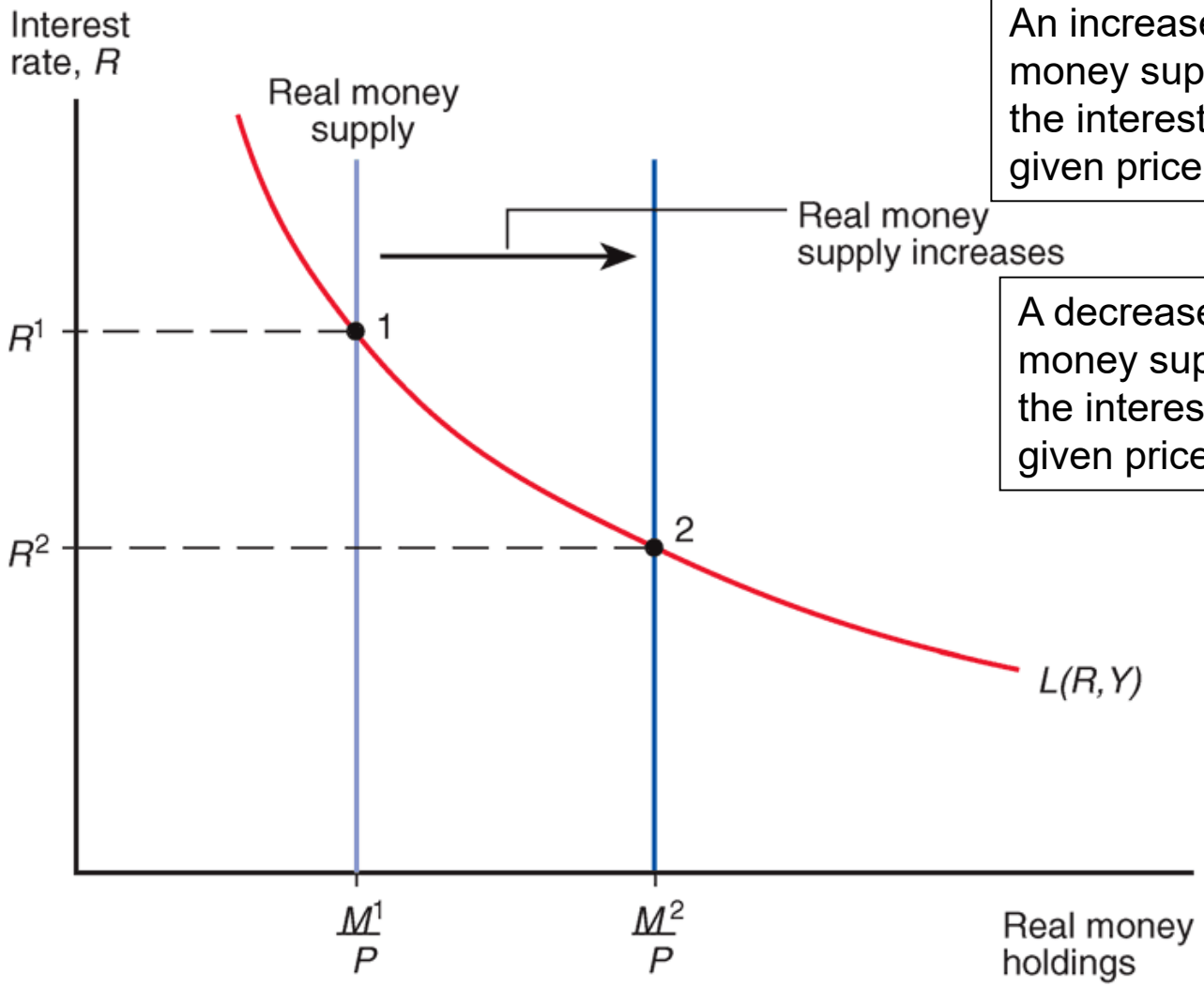


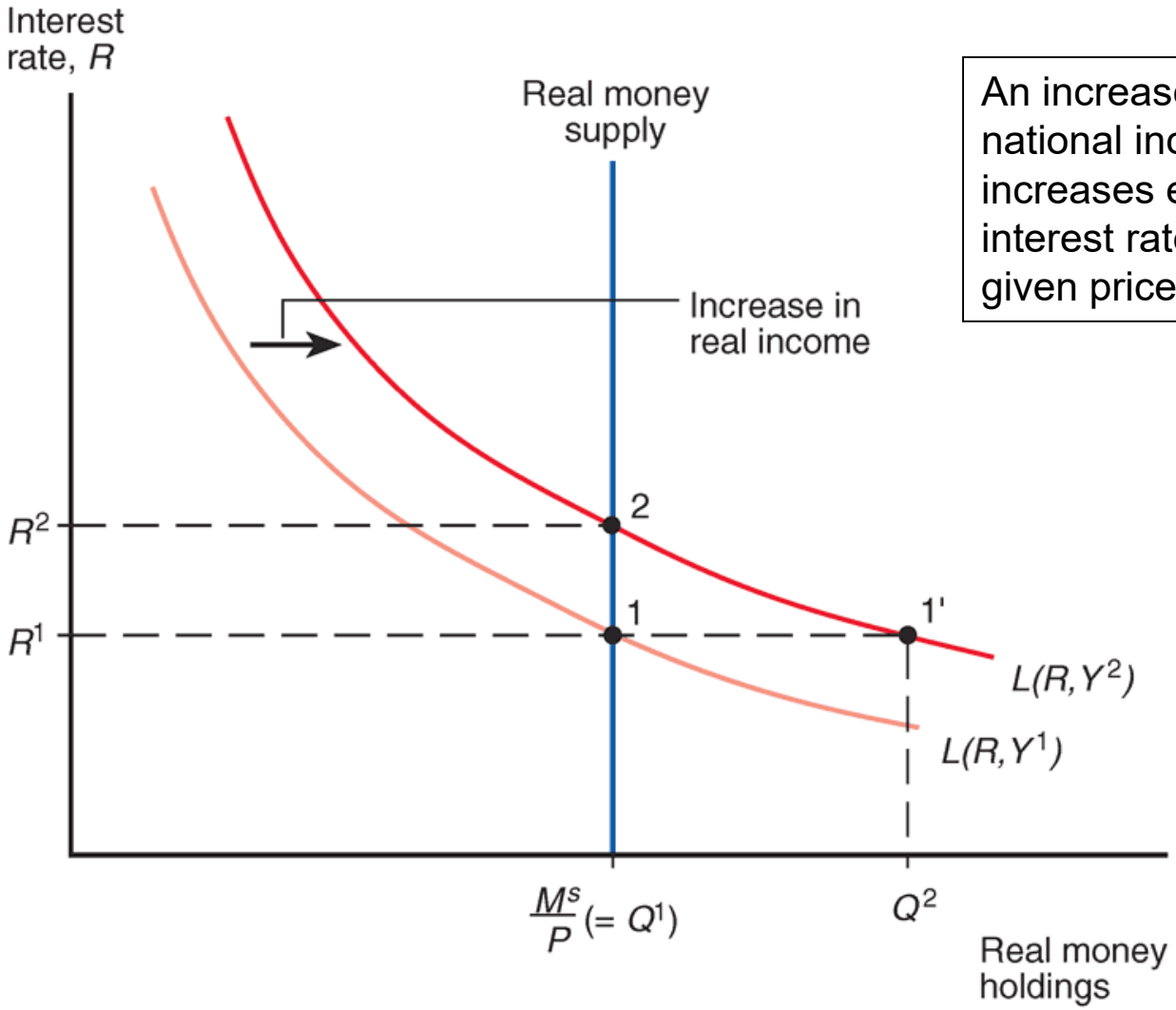
Fig 4: Effect of an Increase in the Money Supply on the Interest Rate



An increase in the money supply lowers the interest rate for a given price level.

A decrease in the money supply raises the interest rate for a given price level.

Fig 5: Effect on the Interest Rate of a Rise in Real Income



An increase in national income increases equilibrium interest rates for a given price level.